Problem Statement No.25 - AI Job Mentor for Informal Workers

1. The Challenge

Millions of informal workers, including electricians, plumbers, carpenters, masons, artisans, mechanics, daily-wage labourers, delivery partners, and self-employed individuals, face significant challenges in finding reliable employment and improving their livelihoods.

Key challenges include:

- **Limited Access to Job Opportunities:** Skilled workers often depend on personal contacts or contractors to find work, with limited access to verified job openings, local employment opportunities, and suitable customers.
- **Lack of Skill Development Guidance:** Many workers are unaware of relevant vocational training programmes, certification courses, and opportunities to upgrade their practical skills.
- **Limited Awareness of Government Schemes:** Workers often lack information about government welfare schemes, financial assistance, skill-development initiatives, and self-employment support.
- **Digital Literacy and Language Barriers:** Complicated job portals, English-centric platforms, and limited digital knowledge make it difficult for rural and semi-urban workers to access employment-related information.
- **Absence of Personalized Career Guidance:** Informal workers lack affordable mentorship to identify suitable work opportunities, improve their skills, and explore better income prospects.

Additionally, fresh graduates, students, and early-career professionals face challenges with resume improvement, interview preparation, and identifying suitable career opportunities.

2. The Objective

The objective of this project is to develop **RozgarMitra AI**, an IBM Granite-powered, multi-agent AI job mentoring platform designed primarily to empower informal workers with accessible employment information, personalized career guidance, and skill-development support.

Built using **IBM watsonx.ai, IBM Granite, and LangFlow**, the platform aims to bridge the employment information gap between workers and available livelihood opportunities.

The solution provides:

1. **Personalized Worker Profiling:** Creates and manages worker profiles based on occupation, practical skills, work experience, qualifications, location, and employment preferences to deliver relevant opportunities.
2. **Skill Development and Training Guidance:** Identifies relevant skill gaps and recommends suitable vocational training programmes, certification courses, and upskilling opportunities to improve employability.

3. **Government Scheme and Financial Awareness:** Provides accessible information about relevant government welfare schemes, financial assistance programmes, self-employment initiatives, and skill-development resources.
4. **Resume and Interview Assistance:** Provides additional support to graduates and formal job seekers through resume analysis, ATS improvement suggestions, role-specific interview questions, and interview preparation.
5. **Domain-Guardrailed Multi-Agent Assistance:** Uses specialized AI agents to handle job guidance, skill development, resume support, and career-related queries while maintaining contextual conversations and restricting irrelevant responses.